Appendix C. Methodological rationale for theory-driven simulation

We design the numerical simulation as a theory-driven computational experiment bridging analytical solvability and complex dynamics. Simulations address non-linear interactions between technological iteration (u_2) and regulatory deterrence (f_p), allowing visualization of trap formation or rupture pathways. This aligns with computational thinking in information systems research, treating the model as a dynamic system for optimization.

Parameter values were calibrated based on multiple credible sources. Fixed and operational costs (C_{p1}, C_{p2}) draw from industry reports on AI content moderation expenses. Creator revenues (E_m, E_u) reflect compliant versus gray-market monetization ranges documented in platform analytics. Government penalties (f_p) are benchmarked against the EU AI Act (Regulation (EU) 2024/1689, Articles 99–101). Moderation efficiencies (u_1, u_2) are grounded in deepfake detection benchmarks from computer vision literature. Full details and sensitivity ranges appear in Section Appendix E.

Parameter values are established through calibration rather than direct empirical estimation, drawing from the following sources. Literature provides foundational ranges for costs and benefits. Industry reports provide monetization models and AI operational costs. Regulations provide benchmarks for f_p and f_{b1} (e.g., EU AI Act). Technological benchmarks provide detection accuracy data informing u_1, u_2. This calibration ensures parameters represent governance regimes: f_p as institutional accountability strength and u_1, u_2 as layered verification capacity.